

PATIENT OUTCOMES · EXPLORATORY DATA ANALYSIS

Patient Readmission Risk Analysis

An Exploratory Data Analysis of 500,000 Hospital Records

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BUSINESS CONTEXT

Why This Matters

01 **Readmissions are costly**

Unplanned returns increase care costs and pressure already-limited capacity.

02 **They signal quality**

Readmission rates are a key metric for evaluating healthcare outcomes.

03 **Many may be preventable**

Earlier identification can support better discharge planning and follow-up.

OUR GOAL

Identify the factors that drive readmission risk — then translate them into actionable recommendations.

[Refine this goal based on your analytical scope]

DATASET OVERVIEW

The Data We Analyzed

PROPERTY	VALUE
Source	Synthetic Healthcare Dataset
Records	500,000
Columns	22
Size	62.3 MB
Target	Readmission Risk

500,000

hospital records

22

columns

62.3

MB

Target variable

Readmission Risk

Preparing the Data

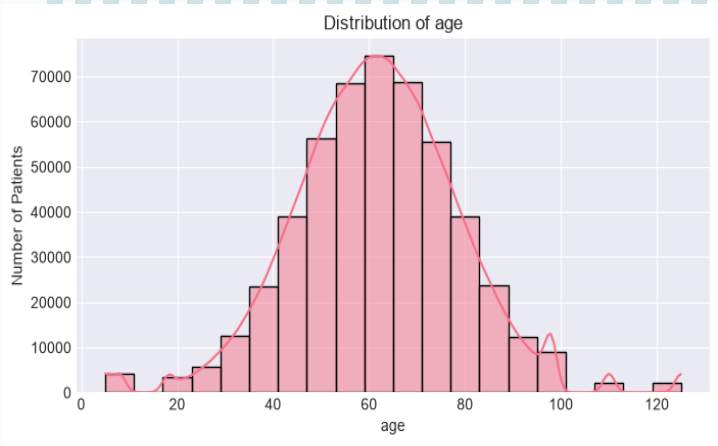
01	**Missing values**	Affected columns: BP, heart rate, temperature, hemoglobin, creatinine and glucose. Handling: Missing values were replaced using the median of each column.
02	**Outliers**	Detection: Identified using IQR (Interquartile Range) and box plots. Treatment: Extreme values were investigated and documented; no values were removed without sufficient evidence
03	**Duplicates**	Duplicate records removed: 0
04	**Standardization**	Checked Gender, Admission Type, Medical Condition, Hospital Department and Insurance Type. No obvious inconsistent values were found, so no standardization changes were required.
05	**Date handling**	Converted Admission Date and Discharge Date from text to datetime format. Used the dates to validate Length of Stay and identified 8,441 mismatches for investigation.

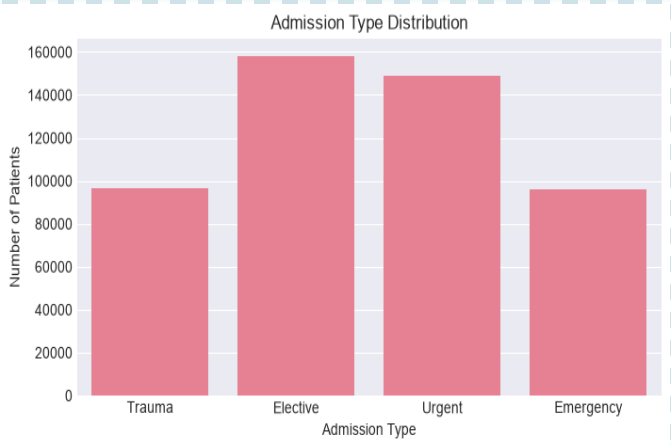
Keep the cleaning log reproducible: record what changed, why, and how many rows were affected.

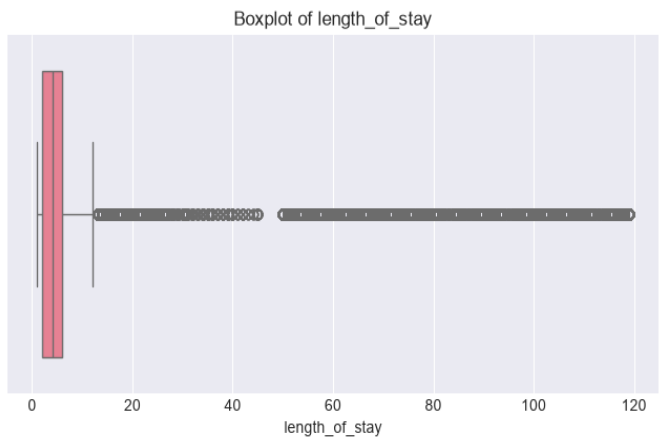
KEY FINDING 1 · UNIVARIATE

What We Learned from Individual Variables

- Finding 1 — Age : The dataset is dominated by older patients, with a mean age of 61.45 years; this highlights the importance of age in understanding the patient population
- Finding 2 — Admission Type: Emergency and Trauma admissions are prominent, showing that a substantial share of patients require urgent care; this matters for hospital resource planning.
- Finding 3 — Length of Stay: The median length of stay is 4 days, with some stays extending up to 119 days; this wide range indicates that some patients require much longer hospital resources







Use one sentence per visual: what happened, how large was it, and why does it matter?

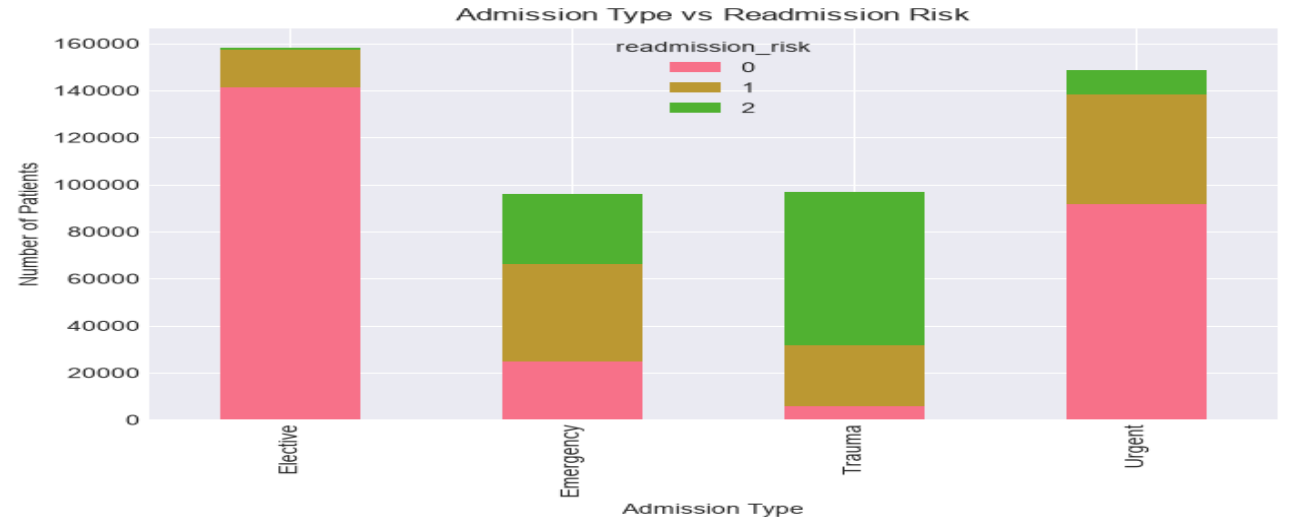
KEY FINDING 2 · BIVARIATE

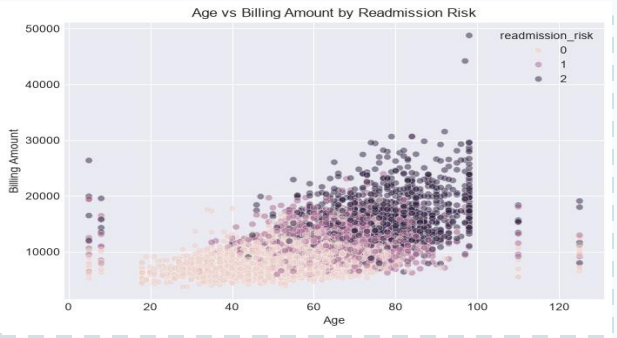
Relationships Between Variables

1. Finding 1 — Readmission risk varies across admission types, with Trauma admissions showing the highest average risk ;this helps identify admission groups needing closer attention

2. Finding 2 — Age and billing amount show a positive relationship, with higher-risk patients generally appearing at higher ages and costs

3. Finding 3 — Higher readmission-risk groups show higher billing amounts, highlighting a subgroup with greater healthcare costs.







KEY FINDING 3 · MULTIVARIATE

What Correlates Most with Readmission Risk?

TOP FACTORS

1 Total Cost

Total cost has the strongest positive correlation with readmission risk(r=0.69),making it the top correlated factor.

2 Creatinine

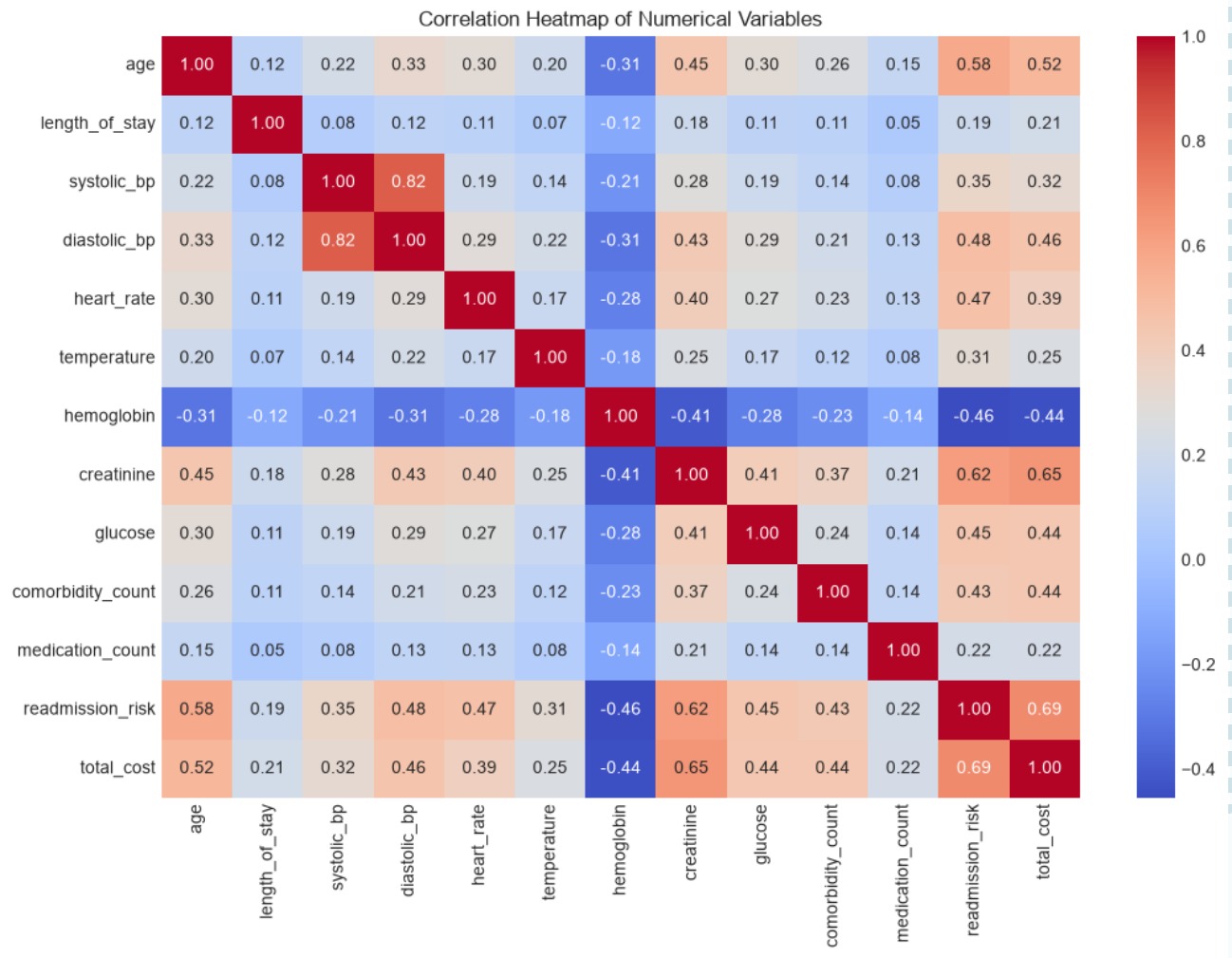
Creatinine has the second-strongest positive correlation(r-0.62), indicating a notable association with readmission risk

3 Age

Age has the third-strongest positive correlation(r=0.58), showing that older patients tend to have higher readmission risk.

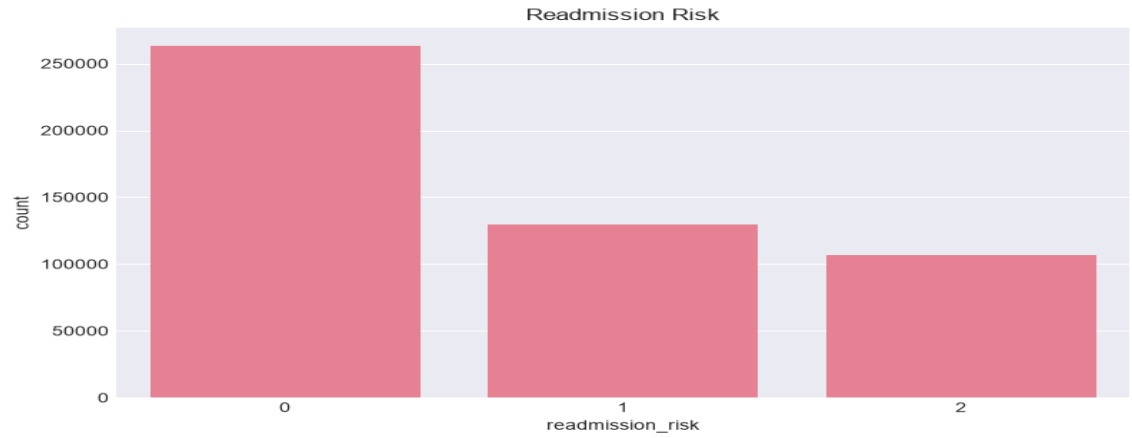
Correlation ≠ causation

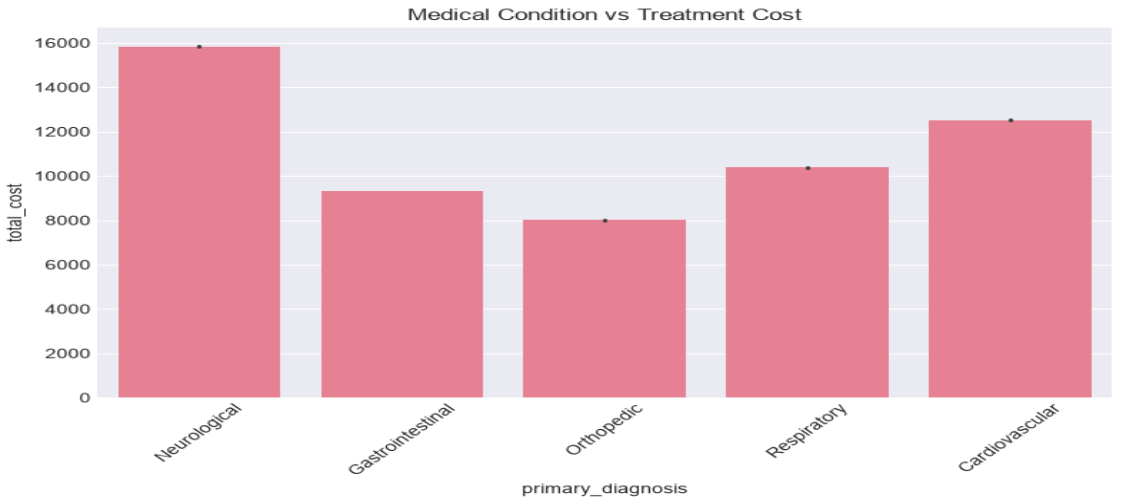
Correlation is not equal to causation. Other factors may influence these relationships, so further validation is needed.

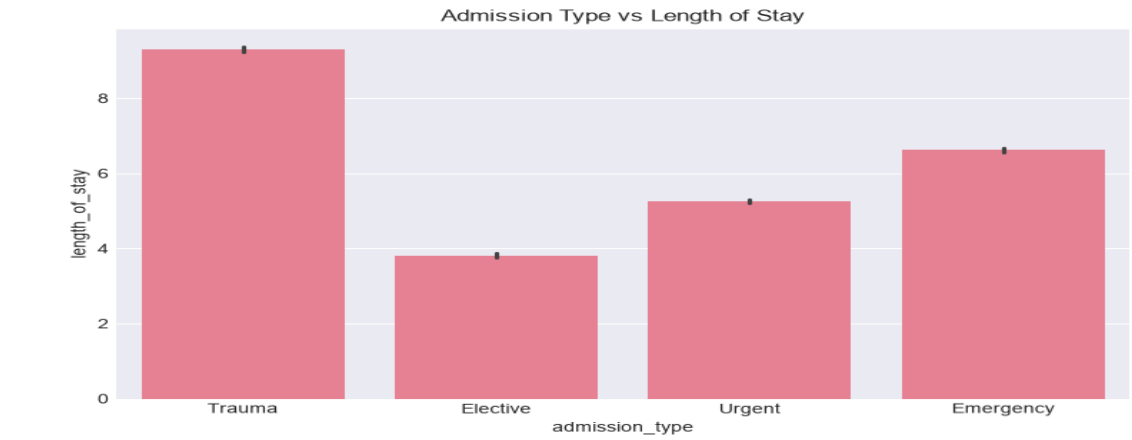


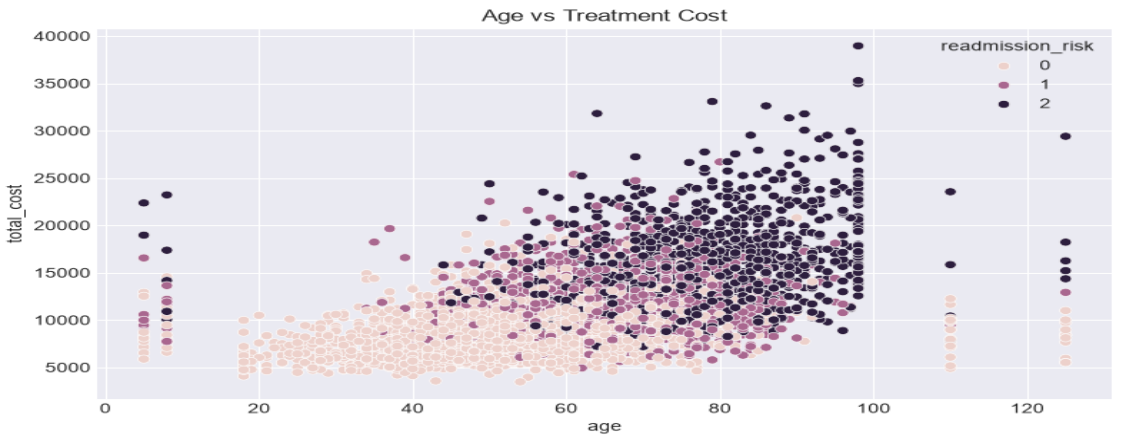
DASHBOARD

All Findings at a Glance









[Optional one-sentence takeaway from the dashboard]

RECOMMENDATIONS

What Should the Hospital Do?

01	**Patient Care**	Recommendation 1 — Provide closer follow-up for older and high-risk patients after discharge.
02	**Resource Allocation**	Recommendation 2 — Allocate additional staff and resources to Trauma and Emergency admissions, which show longer stays and higher risk patterns
03	**Preventive Measures**	Recommendation 3 — Introduce pre-discharge follow-up and monitoring for high-risk patients to reduce potential readmissions.
04	**Cost Optimization**	Recommendation 4 — Review high-cost cases and long hospital stays to identify opportunities to reduce avoidable treatment costs.
05	**Data Collection**	Recommendation 5 — Record previous admissions consistently to improve future readmission-risk analysis and patient-profiling.

Summary

RECAP

- Finding 1- Readmission Risk Total cost, creatinine, and age were the top factors correlated with readmission risk.
- Finding 2- Cost & Conditions
  Neurological conditions had the highest average treatment cost.
- Finding 3- Hospital Stay
  Trauma admissions had the longest average stay at 9.31 days.

IMPACT

Targeted follow-up, better resource allocation, and cost monitoring can support better patient care while improving resources use and controlling healthcare costs.

NEXT STEPS

Validate these findings with additional clinical data and monitor whether the recommended interventions reduce readmissions, length of stay and avoidable costs.

Thank you — Questions?

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